

CMPE491 Senior Project

BioAuth Project Analysis Report

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1. Introduction

In the financial sector, modern digital onboarding and identity verification processes depend a lot on facial recognition and liveness detection. To make sure that the identity being presented matches a real person, these traditional methods usually use visual behavior-based challenges like blinking, turning your head, or repeating random numbers. While these methods have historically been effective against basic spoofing attacks utilizing printed photographs or pre-recorded videos, they are now confronted with an increasingly perilous threat environment.

The rise of advanced generative AI models and tools for animating portraits has changed the risk profile of visual behavior-based liveness detection in a big way. LivePortrait and other similar tools can make very realistic facial animations from just one or a few pictures of a person's face. This includes natural head movements and facial expressions. This feature lets enemies make fake presentations that might be able to pass motion-based liveness tests, which goes against the security assumptions of current authentication pipelines.

The FinCEN 2024 alert made it clear to banks and other financial institutions that deepfake-enabled fraud is becoming more common and is targeting identity verification and authentication processes. This regulatory signal makes it clear that we need to quickly come up with authentication methods that go beyond analyzing visual behavior and include signals that are hard to make or copy.

The BioAuth project fills this important gap by suggesting a biometric authentication system that is aware of physiological liveness. Instead of just looking at facial movements, BioAuth uses remote photoplethysmography (rPPG) technology to find tiny changes in skin color that happen when blood flows under the surface of the face.

Because these physiological signals come from real cardiovascular activity, screens, printed media, and current generation deepfake tools can't copy them. This makes them a much stronger sign of liveness. Our system introduces physiological liveness detection using rPPG signals, which cannot be reproduced by replay attacks or AI-generated portrait animations.

This analysis report gives a full look at the BioAuth system. It includes an evaluation of current systems and their weaknesses, a full list of the functional and non-functional requirements for the proposed system, and system models that include use cases, object models, dynamic behavior models, and user interface design considerations.

2. Current System

Banks, fintech companies, and digital onboarding platforms use remote identity verification systems that usually follow a multi-step pipeline that includes checking documents and biometric liveness checks. A standard authentication process goes like this:

- The user captures an image of their identity document (national ID, passport, driver's license).
- The system extracts facial information from the document using OCR and face detection.
- The user is prompted to perform a selfie or video capture through their device camera.
- Active liveness challenges are presented: the user may be asked to blink, turn their head in specific directions, or repeat randomly displayed digits.
- The system compares the facial embedding from the document with the live capture to establish identity match.
- A pass/fail decision is rendered based on liveness challenge completion and facial similarity scores.

2.1 Limitations of Current Systems

While the architecture described above has been effective against first-generation spoofing attacks, several critical vulnerabilities have emerged:

- **Vulnerability to AI-Generated Content:** Portrait animation tools can generate realistic head movements, blinking, and facial expressions from static images, potentially satisfying motion-based liveness challenges without requiring a live person.
- **Replay Attack Sophistication:** High-resolution screen replays with pre-recorded liveness responses can deceive systems that rely primarily on visual motion analysis.
- **Injection Attacks:** Adversaries can bypass the physical camera entirely by injecting pre-crafted video streams through virtual camera software or modified device drivers, circumventing device-level integrity checks.
- **Single-Modality Weakness:** Systems that depend exclusively on visual cues operate within a single sensory modality, making them fundamentally susceptible to any technology capable of generating visually convincing outputs.

These limitations drive the creation of authentication systems that use physiological signals as an extra layer of verification. This is the main goal of the BioAuth project.

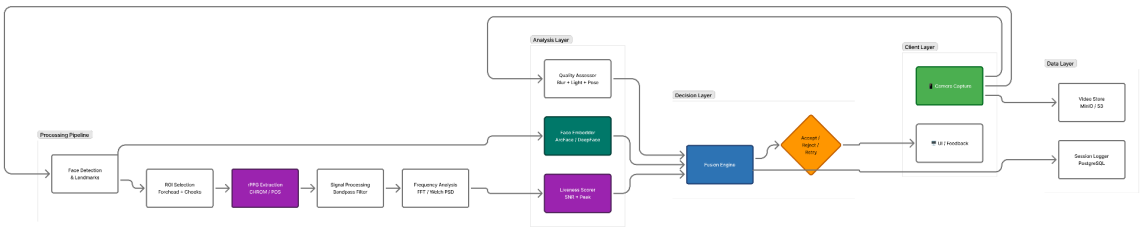
3. Proposed System

3.1 Overview

BioAuth is a hybrid biometric authentication system that enhances traditional facial similarity verification with remote photoplethysmography (rPPG)-based physiological liveness detection. This system is intended to function as an additional security layer within financial identity verification processes, offering protection against presentation attacks that take advantage of visual behavior-based liveness mechanisms.

The primary innovation is found in the extraction and analysis of involuntary physiological signals—namely, the subtle periodic changes in facial skin color resulting from cardiac-driven blood volume fluctuations—as indicators of biological liveness. These signals are captured non-invasively using standard cameras and are processed through signal processing algorithms (CHROM and POS) to extract pulse characteristics. The presence, quality, and physiological plausibility of these signals act as a strong discriminator between live individuals and spoofed presentations.

The system utilizes a multi-factor fusion-based decision engine that combines facial similarity scores, motion/quality validation outcomes, and rPPG-derived liveness scores to generate a final authentication decision, which is classified as Accept, Reject, or Retry.



3.2 Functional Requirements

3.2.1 Face Detection and Tracking (FR-01)

A human face in the camera input will be continuously detected and tracked by the system in real time. To provide adequate temporal resolution for subsequent processing, face detection must run at least 30 frames per second. To avoid ambiguity, the system will handle single-face scenarios and reject multi-face inputs.

3.2.2 Facial Landmark Extraction (FR-02)

MediaPipe Face Landmarker or a comparable framework will be used by the system to extract facial landmarks. For the purpose of extracting the rPPG signal, these landmarks will be used to identify Regions of Interest (ROI) on the face and forehead regions. Over successive frames, landmark tracking must preserve temporal consistency.

3.2.3 Region of Interest (ROI) Selection (FR-03)

Based on identified landmarks, the algorithm will dynamically choose and crop the forehead, left cheek, and right cheek facial ROI regions. To ensure constant signal quality regardless of the user's distance from the camera, ROI dimensions must be adaptively sized in relation to the detected face bounding box.

3.2.4 Physiological Signal Extraction (FR-04)

The system shall extract raw rPPG signals by computing spatially averaged RGB values from the selected ROIs over a temporal window. Both CHROM (Chrominance-based) and POS (Plane Orthogonal to Skin) algorithms shall be implemented for signal extraction. The system shall apply bandpass filtering (0.7–4.0 Hz) to isolate the cardiac frequency band corresponding to 42–240 BPM.

3.2.5 Signal Quality Assessment (FR-05)

The system shall evaluate the quality of the extracted rPPG signal before proceeding with liveness determination. Quality metrics shall include signal-to-noise ratio (SNR), motion artifact magnitude, illumination stability, and temporal signal consistency. Signals failing quality thresholds shall trigger a Retry state rather than a definitive Reject.

3.2.6 Liveness Detection (FR-06)

The system shall perform frequency-domain analysis on the filtered rPPG signal using FFT or Welch Power Spectral Density estimation. Liveness shall be determined by evaluating the presence of a dominant frequency peak within the physiological heart rate range (0.7–4.0 Hz). A liveness score shall be computed based on SNR, peak prominence, spectral concentration, and temporal stability of the detected pulse frequency.

3.2.7 Face Similarity Verification (FR-07)

The system shall extract facial embeddings using deep learning architectures such as ArcFace or DeepFace. Cosine similarity or Euclidean distance shall be computed between the live capture embedding and the stored reference identity embedding. A configurable threshold shall determine identity match acceptance.

3.2.8 Motion and Quality Validation (FR-08)

The system shall assess input quality parameters including blur level, illumination adequacy, face coverage ratio, pose stability, and frame rate consistency. Low-quality inputs shall be flagged with appropriate user feedback messages before authentication processing proceeds.

3.2.9 Multi-Factor Decision Fusion (FR-09)

The system shall integrate results from facial similarity verification, motion/quality validation, and rPPG liveness analysis through a fusion engine. The fusion mechanism may employ rule-based logic or a lightweight machine learning classifier. The final decision shall be one of three states: Accept (live, matched identity), Reject (spoof detected or identity mismatch), or Retry (insufficient signal quality for reliable determination).

3.2.10 Logging and Evaluation (FR-10)

The system shall log all authentication attempts with associated metadata including timestamps, quality scores, liveness scores, similarity scores, fusion decision rationale, and environmental conditions. Logs shall support computation of PAD metrics (APCER, BPCER) per ISO/IEC 30107-3.

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

- The system shall process video input at a minimum of 30 FPS for real-time operation.
- The end-to-end authentication decision (from stable face capture to final decision) shall be rendered within 3–5 seconds.
- The rPPG signal extraction pipeline shall operate with a maximum latency of 100ms per frame on standard consumer hardware.

3.3.2 Usability Requirements

- The system shall provide real-time visual feedback including camera preview, face detection overlay, and contextual guidance messages (e.g., 'Please improve lighting', 'Hold steady').
- Authentication results shall include brief explanations for Reject and Retry outcomes.
- The interaction shall be non-intrusive, requiring no complex or unnatural user actions beyond facing the camera.

3.3.3 Security and Privacy Requirements

- All biometric data shall be processed in volatile memory and shall not be persistently stored without explicit user consent.
- The system shall comply with GDPR and KVKK data protection principles.
- The system shall detect and resist print, replay, and AI-generated portrait animation attacks through rPPG-based physiological liveness verification.
- No biometric data shall be transmitted over the network without encryption.

3.3.4 Reliability and Availability Requirements

- The system shall maintain stable operation without crashes during continuous use sessions.
- The system shall gracefully degrade under adverse conditions (low light, excessive motion) by issuing Retry rather than incorrect Reject decisions.

3.3.5 Portability and Deployment Requirements

- The system shall support deployment on web platforms (React + WebRTC) and Android mobile devices.
- The system shall operate with standard webcams and smartphone cameras without requiring specialized hardware.
- The backend shall be deployable as a containerized service using FastAPI.

3.4 Pseudo Requirements

- The backend shall be implemented in Python using OpenCV, NumPy, SciPy, and MediaPipe.
- The web frontend shall be built with React and utilize WebRTC for camera access.
- The mobile prototype shall target Android (Kotlin) with native camera integration.
- The API layer shall use FastAPI with WebSocket support for real-time communication.
- PostgreSQL shall be used for logging and metric storage; MinIO/S3 for test video recordings.
- The project shall use Git for version control with a structured branching strategy.

3.5 System Models

3.5.1 Scenarios

Scenario 0 – Normal User Authentication: On their smartphone, a genuine user launches the BioAuth app. When there is enough light, they put their face in front of the camera. After identifying their face and extracting landmarks, the system starts recording rPPG signals from ROIs in their cheek and forehead. CHROM/POS algorithms extract a distinct pulse signal with SNR above the acceptance threshold after three to four seconds of steady capture. A high similarity score is obtained by comparing the facial embedding to the reference identity. After combining all the variables, the fusion engine produces Accept along with a confidence score and detected BPM.

Scenario 1 – Print Attack: A printed photo is held in front of the camera by an assailant. After identifying a face, the machine extracts landmarks. The rPPG signal is flat and lacks a noticeable frequency peak, but ROI analysis shows no temporal fluctuation in RGB channels. The minimum criteria is not met by the liveness score. Reject is output by the system with the reason code "No physiological signal detected."

Scenario 2 – Replay Attack: On a phone screen placed in front of the camera, an attacker plays a pre-recorded video of the authorized user. The rPPG signal derived from the screen replay shows typical artifacts, such as poorer SNR compared to live tissue, irregular frequency peaks due to display refresh rate interference, and reduced signal amplitude due to screen pixel discretization, even though facial likeness may match. The signal quality abnormality is flagged by the system, which then produces Reject.

Scenario 3 – AI-Generated Portrait Animation Attack: An attacker creates realistic head movements and facial expressions from a stolen photo using a portrait animation application (like LivePortrait). Visual motion patterns are successfully mimicked by the created animation. However, because there is no real blood circulation underneath the pixel values, the synthesized video lacks true physiological micro-color fluctuations. Either noise or a non-physiological signal pattern is extracted by the rPPG pipeline. With the reason code "Physiological signal inconsistent," the system outputs Reject.

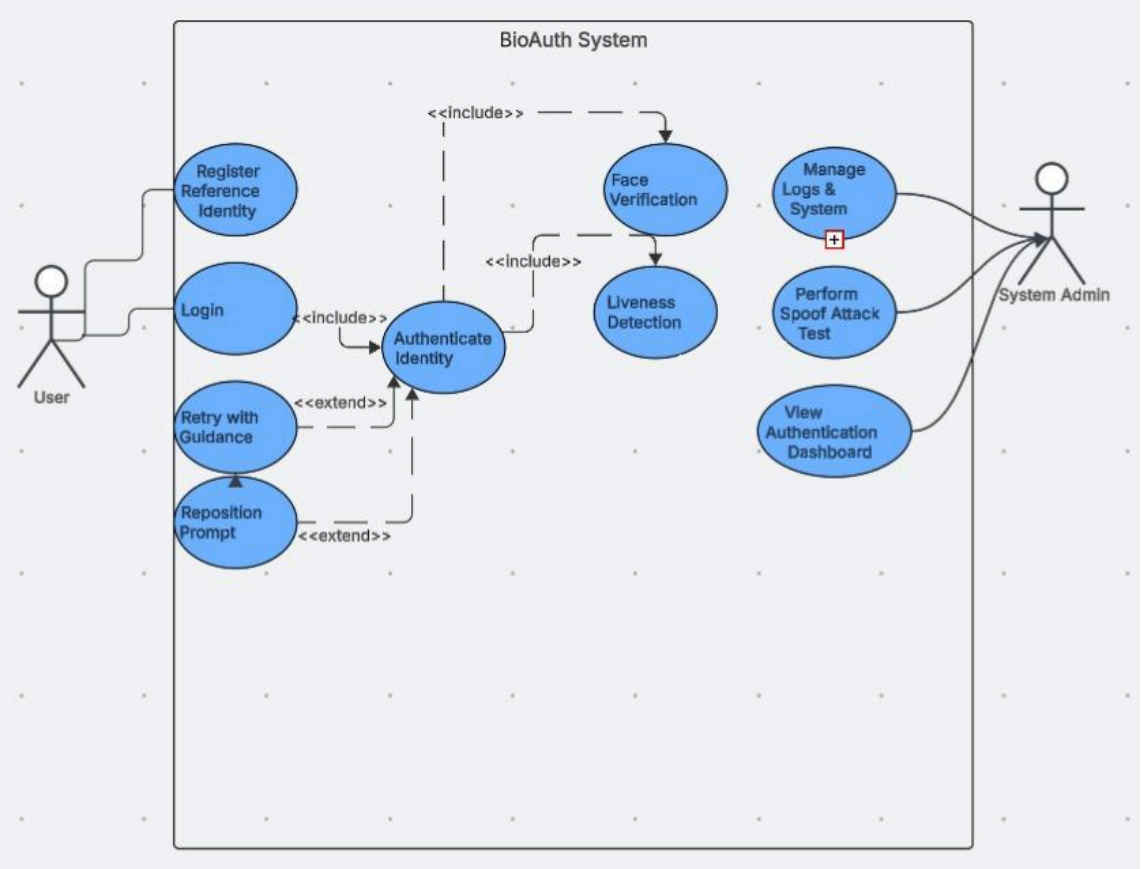
Scenario 4 – Low Quality / Environmental Degradation: In a poorly light setting, a legitimate user tries to authenticate. The quality assessment module finds that there is not enough light for accurate rPPG extraction even though the algorithm recognizes a face. Instead of generating an inaccurate liveness evaluation, the system generates The user is prompted to modify their surroundings with the guidance notice "Please improve lighting conditions" when they try again.

3.5.2 Use Case Model

The following use cases describe the primary interactions between actors and the BioAuth system:

Actors:

- User: The individual attempting to authenticate their identity.
- System Administrator: Manages system configuration, reviews logs, and adjusts thresholds.
- Backend Service: Processes authentication requests, executes the rPPG pipeline, and returns decisions.



Use Case ID	UC-01
Use Case Name	Authenticate Identity
Actor(s)	User
Precondition	User has a registered reference facial identity in the system.
Main Flow	1. User opens the application and grants camera access. 2. System activates camera and displays live preview. 3. System detects face and begins landmark extraction. 4. System selects ROIs and starts rPPG signal capture. 5. System processes signal via CHROM/POS algorithms. 6. System performs frequency analysis and computes liveness score. 7. System computes facial similarity against reference. 8. Fusion engine integrates all scores and outputs decision.
Postcondition	Authentication result (Accept/Reject/Retry) is displayed to user. Session is logged.
Alternative Flow	If signal quality is insufficient, system outputs Retry with guidance. If face is not detected, system prompts user to reposition.

Use Case ID	UC-02
Use Case Name	View Authentication Dashboard
Actor(s)	System Administrator
Precondition	Administrator has valid access credentials.
Main Flow	1. Administrator logs into the dashboard. 2. System displays recent authentication logs. 3. Administrator filters by date, result type, or quality metrics. 4. System displays rPPG signal visualizations and score distributions. 5. Administrator reviews APCER/BPCER metrics.
Postcondition	Dashboard data is displayed. No data is modified.
Alternative Flow	If no logs exist for the selected period, system displays empty state.

Use Case ID	UC-03
Use Case Name	Register Reference Identity
Actor(s)	User
Precondition	User has not previously registered.
Main Flow	1. User opens registration flow. 2. System captures facial image under controlled conditions. 3. System extracts facial embedding using ArcFace/DeepFace. 4. System stores the reference embedding securely. 5. System confirms successful registration.
Postcondition	Reference identity is stored and available for future authentication.
Alternative Flow	If capture quality is insufficient, system requests recapture.

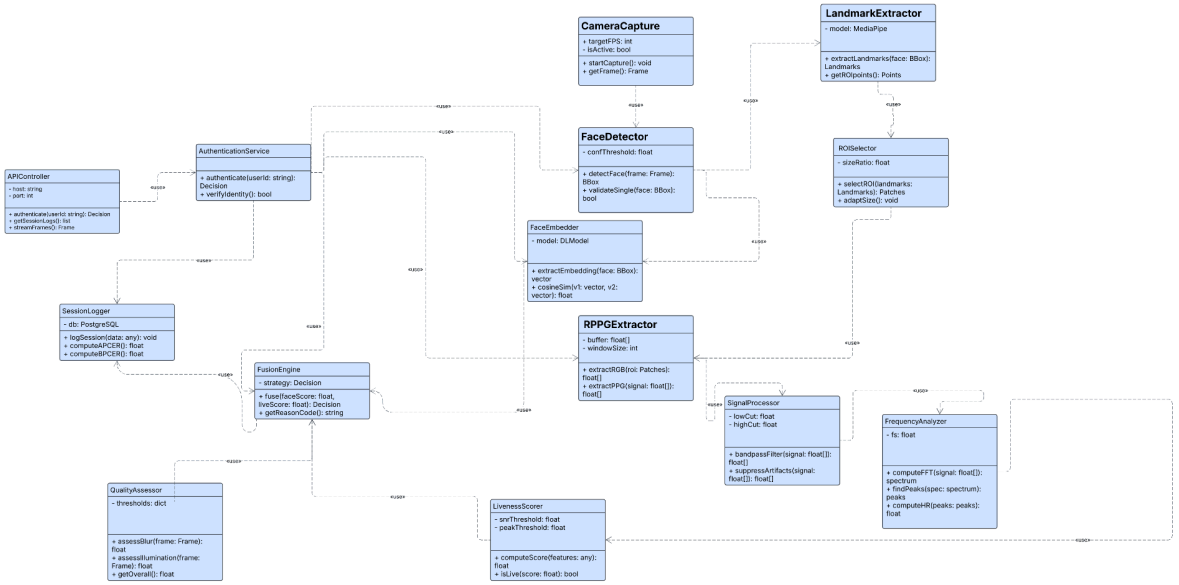
Use Case ID	UC-04
Use Case Name	Perform Spoof Attack Test
Actor(s)	System Administrator
Precondition	Test environment is configured with known attack datasets.
Main Flow	1. Administrator selects attack type (print/replay/animated). 2. System processes attack input through standard pipeline. 3. System logs all intermediate scores and final decision. 4. Administrator reviews detection performance metrics.
Postcondition	Attack detection results are logged with associated metrics.
Alternative Flow	If input format is unsupported, system reports error.

Note:

Some supporting functionalities such as *Login*, *Retry with Guidance*, *Reposition Prompt*, *Face Verification*, and *Liveness Detection* are represented in the use case diagram as **included or extended use cases**. These elements represent internal system processes or supporting interactions and therefore are not described as separate use case tables in this document. Only the primary user-level interactions are documented in the following use case descriptions.

3.5.3 Object and Class Model

The BioAuth system is decomposed into the following principal classes and their responsibilities:



- **CameraCapture:** Manages camera initialization, frame acquisition at the target FPS, and frame buffering. Provides a stream of raw video frames to downstream modules.
- **FaceDetector:** Detects human faces in each frame using MediaPipe or equivalent. Returns bounding box coordinates, confidence score, and face count. Enforces single-face validation.
- **LandmarkExtractor:** Extracts facial landmark points from detected face regions. Identifies key anatomical points used for ROI definition (forehead boundaries, cheek boundaries, eye corners).
- **ROISelector:** Computes ROI coordinates based on landmark positions. Dynamically adjusts ROI dimensions relative to face size. Outputs cropped ROI image patches for each frame.
- **RPPGExtractor:** Computes spatially averaged RGB values from ROI patches. Implements CHROM and POS algorithms for rPPG signal derivation. Maintains a temporal signal buffer.
- **SignalProcessor:** Applies bandpass filtering (Butterworth, 0.7–4.0 Hz) to raw rPPG signals. Performs motion artifact suppression. Outputs filtered physiological signal time series.
- **FrequencyAnalyzer:** Performs FFT or Welch PSD on filtered signals. Identifies dominant frequency peak and computes SNR, spectral concentration, and peak prominence metrics.

- **LivenessScorer:** Evaluates physiological plausibility based on frequency analysis results. Computes a normalized liveness score (0.0–1.0) based on configurable weighting of SNR, peak quality, and temporal consistency.
- **FaceEmbedder:** Extracts 512-dimensional facial embeddings using ArcFace or DeepFace. Computes cosine similarity against reference embeddings. Returns similarity score.
- **QualityAssessor:** Evaluates frame-level and sequence-level quality: blur, illumination, face coverage, pose angle, and FPS consistency. Flags low-quality inputs.
- **FusionEngine:** Integrates liveness score, similarity score, quality score, and motion validation results. Applies decision rules or ML classifier to produce final Accept/Reject/Retry decision with confidence and reason codes.
- **SessionLogger:** Records all session metadata, intermediate scores, and final decisions to PostgreSQL. Supports metric computation and audit trail generation.
- **APIController:** FastAPI-based REST/WebSocket endpoint. Manages authentication session lifecycle, receives video frames, returns real-time status updates and final decisions.

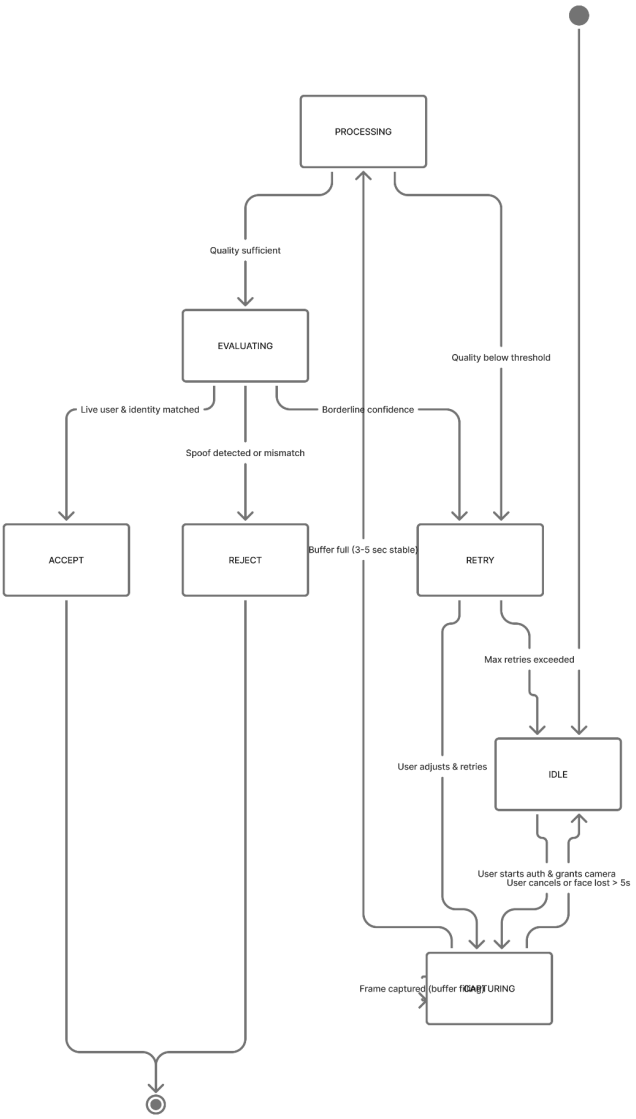
3.5.4 Dynamic Models

The authentication process follows a sequential pipeline with conditional branching based on quality and liveness assessments. The dynamic behavior of the system is described through the following state transitions and activity flows:

Authentication State Machine:

The system operates through the following states during an authentication session:

- IDLE: System initialized, waiting for user to begin authentication.
- CAPTURING: Camera active, face detected, frames being accumulated for rPPG extraction.
- PROCESSING: Sufficient frames collected; rPPG signal extraction, filtering, and frequency analysis in progress.
- EVALUATING: Liveness score, face similarity score, and quality score computed; fusion engine determining final decision.
- ACCEPT: Authentication successful—live user with matched identity confirmed.
- REJECT: Authentication failed—spoof detected or identity mismatch.
- RETRY: Signal quality insufficient for reliable determination; user prompted to adjust conditions.



Activity Flow:

1. User initiates authentication → System transitions from IDLE to CAPTURING.

2. Face detection confirms presence → ROI selection begins, rPPG signal buffer starts filling.

3. After 3–5 seconds of stable capture → System transitions to PROCESSING.

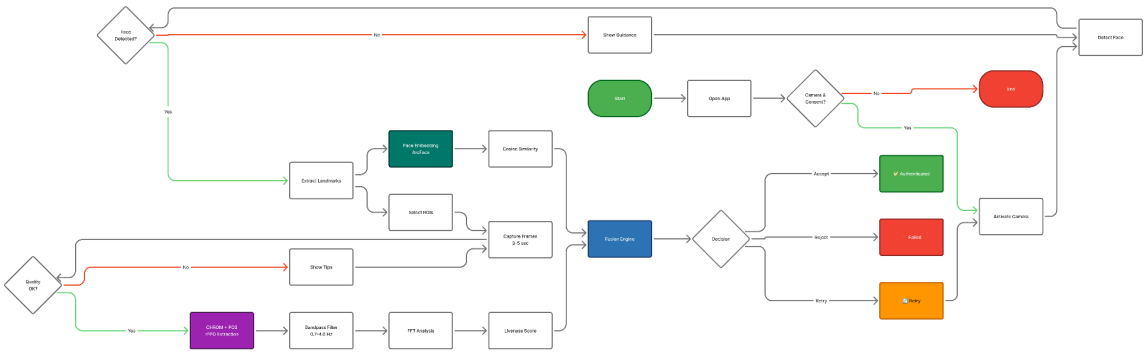
4. CHROM and POS algorithms process buffered signals → Bandpass filtering applied → FFT/Welch analysis performed.

5. Quality assessment runs in parallel → If quality below threshold → RETRY with user guidance.

6. If quality sufficient → System transitions to EVALUATING.

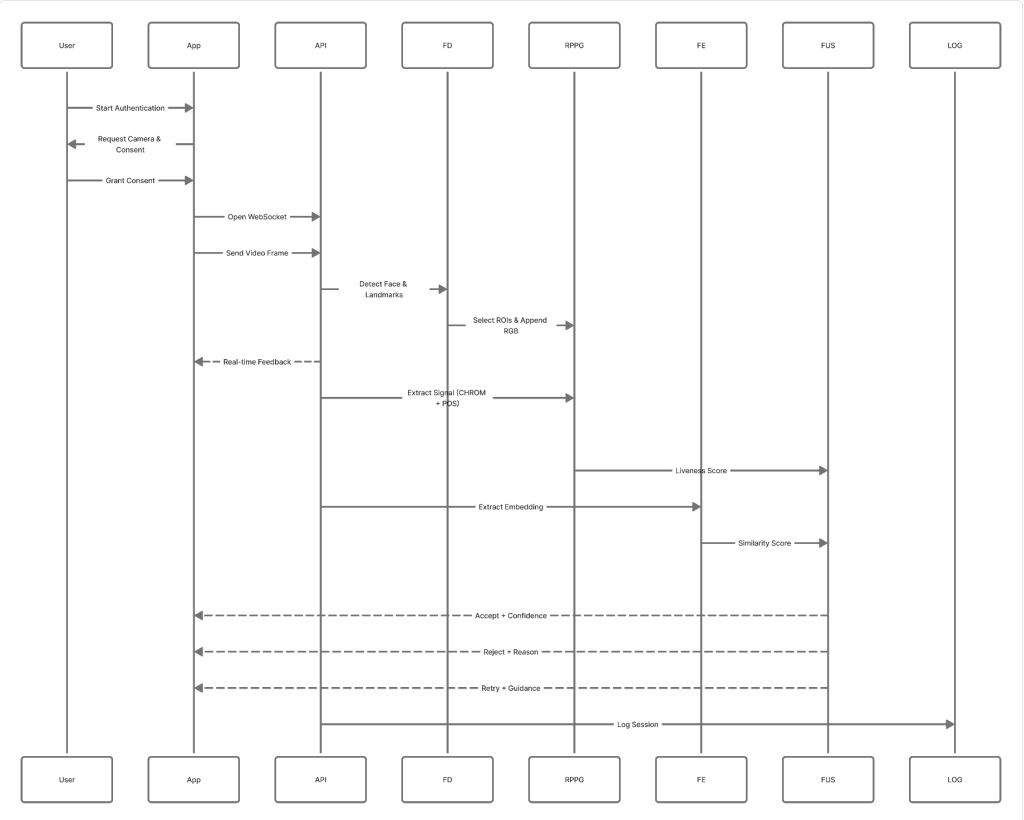
7. Fusion engine integrates liveness score + similarity score + quality score → Final decision computed.

8. System transitions to ACCEPT, REJECT, or RETRY → Result displayed to user → Session logged.



Sequence of rPPG Processing (per frame):

CameraCapture → FaceDetector → LandmarkExtractor → ROISelector → RPPGExtractor (append to buffer) → [When buffer full] → SignalProcessor → FrequencyAnalyzer → LivenessScorer → FusionEngine.



3.5.5 User Interface – Navigational Paths and Screen Mock-ups

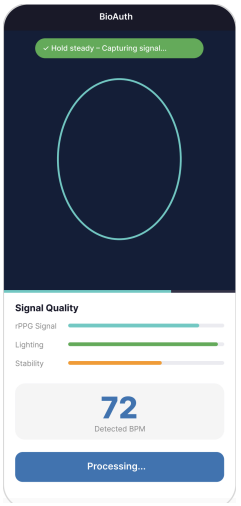
The BioAuth system provides two primary interfaces: a mobile/web authentication interface for end users and an administrative dashboard for system operators.

Authentication Interface (Mobile/Web):

The authentication interface follows a linear flow designed for minimal friction:

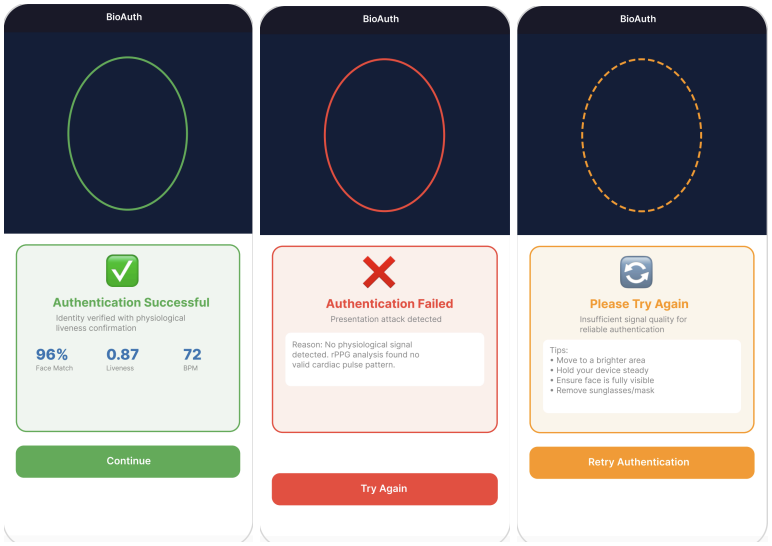
- Screen 1 – Welcome/Consent: Shows a quick description of the biometric authentication procedure and asks for the user's express authorization before using the camera and biometric data. To move on to the next screen, simply click the "Begin Authentication" button.

- Screen 2 – Camera Capture: Full-screen camera preview with a face positioning guide overlay (oval outline). Real-time feedback messages appear at the top of the screen (e.g., 'Position your face within the oval', 'Lighting is too low – please move to a brighter area', 'Hold steady – capturing signal...'). A progress indicator shows the signal capture progress.



- Screen 3 – Processing: Brief animation while the fusion engine computes the final decision. Duration: 1–2 seconds.

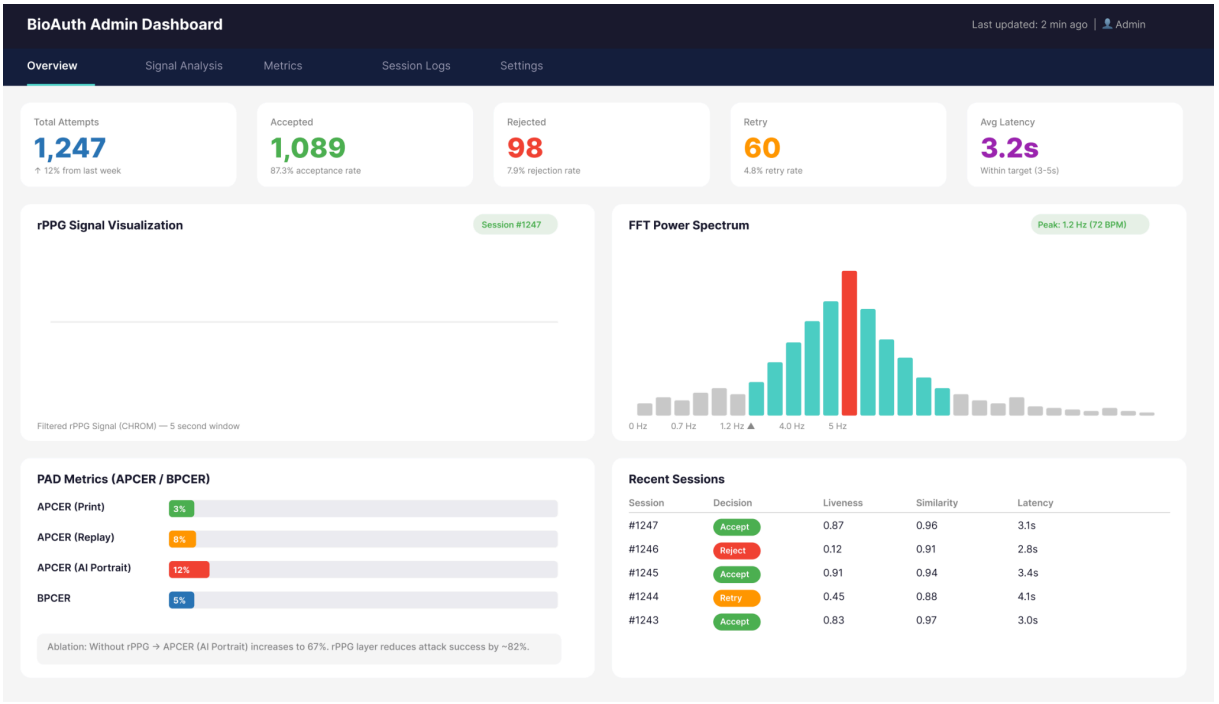
- Screen 4 – Result: Displays the authentication outcome (Accept/Reject/Retry) with a color-coded visual indicator (green/red/yellow). For Reject and Retry, a brief explanation is provided. A 'Try Again' button is available for Retry outcomes.



Administrative Dashboard (Web):

The dashboard provides system monitoring and evaluation capabilities:

- Overview Panel: Summary statistics including total authentication attempts, acceptance rate, rejection rate, retry rate, and average latency.
- Signal Visualization: For selected sessions, displays the raw rPPG signal, filtered signal, FFT spectrum, detected peak frequency, and SNR value.
- Metrics Panel: Computed APCER and BPCER values across different attack types. Configurable date range and filtering options.
- Log Browser: Searchable table of all authentication sessions with sortable columns for timestamp, decision, liveness score, similarity score, quality score, and environment conditions.



4. Glossary

Term	Definition
rPPG	Remote Photoplethysmography – a non-contact method for detecting blood volume pulse signals from facial video by analyzing subtle skin color changes.
CHROM	Chrominance-based rPPG extraction algorithm that uses a linear combination of chrominance signals to isolate the pulse signal.
POS	Plane Orthogonal to Skin – an rPPG extraction method that projects RGB signals onto a plane orthogonal to the skin tone direction.
ROI	Region of Interest – specific facial areas (forehead, cheeks) selected for physiological signal extraction.
SNR	Signal-to-Noise Ratio – a measure of the strength of the desired pulse signal relative to background noise.
FFT	Fast Fourier Transform – an algorithm for converting time-domain signals to frequency-domain representation.
BPM	Beats Per Minute – the unit of heart rate measurement.
APCER	Attack Presentation Classification Error Rate – the proportion of attack presentations incorrectly classified as genuine.
BPCER	Bona Fide Presentation Classification Error Rate – the proportion of genuine presentations incorrectly classified as attacks.
PAD	Presentation Attack Detection – the process of determining whether a biometric sample is a genuine or attack presentation.
ArcFace	A deep learning-based facial recognition model that produces discriminative facial embeddings using additive angular margin loss.
DeepFace	A facial recognition and analysis framework that provides face verification, detection, and embedding extraction capabilities.
MediaPipe	Google's cross-platform framework for building multimodal ML pipelines, used for real-time face detection and landmark extraction.
Deepfake	AI-generated synthetic media that replaces or manipulates a person's likeness in video or images.
Liveness Detection	The process of determining whether a biometric sample originates from a living person rather than a spoof artifact.

Fusion Engine	A decision-making component that integrates multiple independent scores to produce a unified authentication verdict.
KVKK	Kişisel Verilerin Korunması Kanunu – Turkey's Personal Data Protection Law.
GDPR	General Data Protection Regulation – EU regulation on data protection and privacy.

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